

AI Voice Support Agent - Jobbie AI

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Code repo: <https://github.com/NavjeetSinghKukreja/JobbieAI>

Part 1 — Diagnosis

1. What may be causing these issues

The assistant sometimes repeats itself

If we're dumping the full conversation history in the prompt without any duplication or loop check, the model can regenerate simpler phrasing especially if the new question is close to the original question.

We can investigate this by seeing logs of conversations where users complained or dropped off and checking if repetitions correlate with conversation length

Responses become less helpful during long conversations

This is because of the context window as LLMs have a fixed window of tokens, when the conversation grows too large it truncates older messages which leads to losing context and forgets stuff that the user has mentioned earlier

This can be investigated by checking the token counts per request and how close we are to hitting the context limit

The AI occasionally gives incorrect but confident answers

This is caused by hallucination. It happens because the model generates text that's likely even if its not actually verified to fill gaps with plausible sounding information

We can investigate this by checking sample responses and fact-checking them against the actual knowledge base and tracking which topics generate the most hallucinations

Different users get inconsistent responses to similar questions

This is caused due to temperature settings as if the temperature is above 0, the output is non-deterministic and gives a different output for the same input. Even if two users ask the same question their conversation histories are different which changes the prompt

This can be investigated by checking the temperature and sampling settings in production

The assistant occasionally loses track of the conversation topic.

This ties back to the context window management as without explicit topic tracking the model relies on conversation context to understand the user's question.

This can be investigated by reviewing conversations where the user mentions things like "this is not what i asked for" or " I already told you"

Costs increase significantly during longer sessions

As LLM APIs charge per token and each API call sends the full conversation history so the cost grows with every message in the conversation unless we use summarisation.

This can be investigated by tracking token usage per conversation over time

2. Which issues you would prioritise first and why

The two main issues that I would prioritise are

Hallucination: This directly damages user's trust as a support agent that gives wrong information is worse than no agent at all and it could cause real harm if the users act on bad advice

Cost Explosion: This directly impacts the financial aspect of the company and causes it to be unsustainable.

3. What you would monitor in production

The key metrics I would track are:

Quality metrics:

1. Hallucination Rate
2. Repetition Rate
3. Escalation Rate (how often does the system hand off to a human)

Operational metrics:

1. Tokens per request
2. Cost per conversation

User Experience metrics:

1. Conversation satisfaction score
2. Resolution rate

Part 2 — Improved Design

Architecture Improvements

Context window management: Instead of stuffing the entire conversation history into every call we can use a sliding window and combine it with a summarization strategy. Recent messages go in exactly how the user says it whereas the older ones get summarized

RAG (Retrieval-Augmented Generation): Instead of making the assistant answer from memory alone it should retrieve relevant knowledge from base articles and inject them as context. This decreases hallucination and inconsistency

Loop Detection: Before sending a new response, it should be compared against recent responses to avoid repetition. If similarity is too high then the responses should either be rephrased or escalated.

Response Validation: After the LLM generates a response but before sending it to the user, we run it through a validation layer that checks for repetition, relevance to the user's actual question, and format.

Confidence/Escalation Layer: Have the model self-assess confidence and if the confidence is below a threshold it should escalate to a human agent.

Response Cache for Common questions: For frequently asked questions which can be identified through clustering we can cache validated responses. When a new question matches a cached one closely enough we can serve the cache response instead of generating a new one. This also reduces inconsistency and helps save costs.

Tradeoffs:

These layers add latency and complexity. Summarisation is lossy by nature so older context loses detail, RAG is only as good as the knowledge base it pulls from, and response validation adds processing time before every reply. The goal is finding the right balance between reliability and speed, not eliminating all risk.

